

Information provision and network externalities: the impact of genomic testing on the dairy industry

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I made this note
on capitulation last time

Abstract

PLEASE RE-READ

We develop a synthetic control group to estimate the long-term impacts of introducing Genomic Selection in the United States market for dairy cattle genetics. Genomic selection is an application of big data that uses the entire genome of an animal to test for the presence of a set of traits. Unlike pre-existing technologies that require several years of data from a bull's daughters, an animal can be genomically tested as soon as it is born, allowing breeders to identify the "best" animals at a much faster rate. Using a data set of all bulls marketed in the US and Canada from 2000 to 2020, we find that genomic selection significantly increased genetic gains for all measured traits, particularly milk production, protein, and fat yields. It is also the source of increased levels of inbreeding depression, a decrease in the performance of animals whose parents have a high degree of relatedness, as a consequence of genetics companies breeding more animals from established lines rather than lesser-known lines with lower trait values but less inbred. We argue that this is a network externality inflicted by genetics companies upon each other. We can think of the American dairy cattle gene pool as a public resource that animal breeders can abuse by drawing too much from specific combinations of genes whose frequency will increase, leading to potentially disastrous consequences. Our estimation shows that the increased inbreeding rate of American bulls caused a loss of between 2 and 5 billion dollars per year to the entire industry. Solving this externality will require either a mechanism to internalize the harmful effects, such as paying a much higher price for more inbred sires, or a collective action mechanism to select which lines will be bred in the next generation.

max
250
words

Where is
economics
in this abstract?

1 Introduction

This is not the intro of an econ paper

This article studies the consequences of introducing genomic selection in the dairy industry. Genomic selection uses data from an animal's entire set of genes (its genotype) to identify which traits are associated with specific combinations of genes as soon as the animal is born. The most critical consequence of genomic selection is a significant increase in the rate of genetic gain, defined as the intergenerational variation in the quantities of a trait. In other words, every new generation of bulls has even better productivity than the previous, and this difference is growing faster.

Improved genetics via the selection of bulls (sires) and cows (dams) is a crucial driver for productivity improvements in the dairy industry. Milk production in the United States has tripled in the last forty years, and over half of this increase is solely due to genetics¹. The United States is also the world's largest exporter of bovine genetics, with a share of 46.4% of the total value of exports in 2019. Every improvement in genetics in the United States will be quickly transmitted to the rest of the world dairy industry.

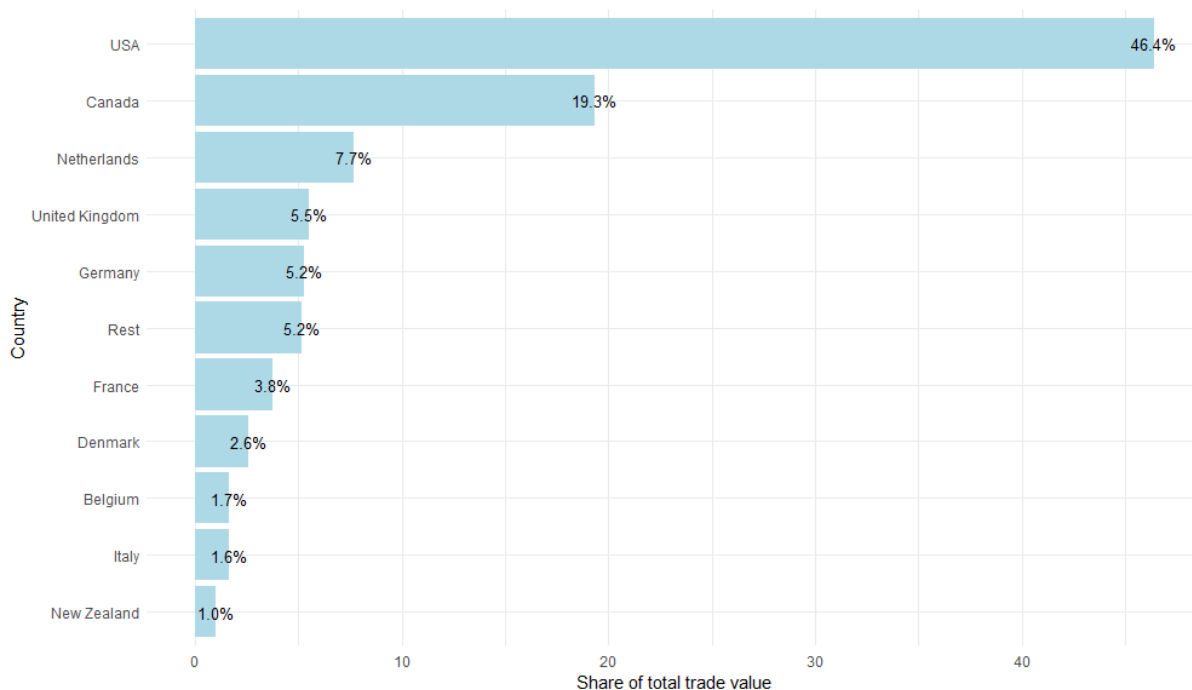


Figure 1: Share of trade Value of Bovine Semen exports by country (2019)

Source: UN Comtrade database

This article will investigate the consequences of a technological innovation: Genomic Selection for bulls. Since the entire genome of the *Bos Taurus* was sequenced in 2007, genetics com-

¹<https://uscdcb.com/impact/>

in-text, not footnote

panies can identify which bulls have the highest value among a set of quantitative traits and can pass them to their offspring.

Increased genetic gains result from shorter generation intervals; breeders can identify a highly productive bull before it reaches sexual maturity. Consequently, inferior sires are “pulled out” of the reproductive pool, leading to higher average levels of all genetic traits. But this mechanism has an additional consequence: increased competition in the animal genetics industry firms that supply bull genetics are now under higher pressure to market more “efficient” bulls. However, the most common way to breed a highly productive sire is through linebreeding, defined as mating two related animals that are themselves descendants of a “celebrity” sire; consequently, the progeny of those animals will have a higher inbreeding rate than their parents. *they're not called this*

Traditional selection techniques relied on information from pedigrees and observed trait values to assess an animal's fitness. Genomic selection uses data from an animal's genome to accurately predict breeding values, particularly in cases where traits are expressed only in females. In such case, it was necessary to measure the traits of a large number of daughters from the same bull; for that to happen, these cows must have reached the age of sexual maturity (one year); hence, the entire process could take three to four years until enough information was gathered.

Instead, genomic selection uses the bull's DNA to detect gene combinations significantly different than the expected frequency if they were associated at random (Goddard and Hayes 2007) to detect the presence of certain genetic traits that are highly correlated with specific gene combinations. This process can be undertaken as soon as the calf is born, so by the time the animal has reached sexual maturity (one year old), it can be marketed immediately.

Before the bovine genome was sequenced, it was believed that genomic testing would help animal breeders to identify the best sires; nevertheless, genomic testing had the opposite effect. For example, Daetwyler et al. (2007) argued that genomic selection would reduce inbreeding since it allows for a more accurate estimation of breeding values. This greater accuracy would enable breeders to “discover” new potentially high-performing sires independent from their ancestry; the data shows that precisely the opposite happened after 2009.

The introduction of genomic testing has also led to a higher prevalence of inbreeding depression ~~attributable to a higher frequency of recessive genes due to the relatedness of an animal's parents~~. This results from genetics companies supplying more bulls from “prestigious,” more inbred pedigrees rather than lesser-known but less inbred alternative pedigrees. Unfortunately, they do not consider that the average inbreeding rate across all animals has steadily increased since 2010 (Cole 2019).

In economic terms, inbreeding depression can be understood as a negative externality that genetics companies inflict upon each other because of breeding more closely related bulls. In this case, the profits of genetics companies are a function of the trait levels of the animals they produce but also depend on the degree of relatedness of bulls to each other. The more genetic companies generate into a family of animals (also known as a "line"), the higher the likelihood of observing inbreeding depression in the next generation.

This is an example of a **network externality** (Liebowitz and Margolis 1994), (Katz and Shapiro 1985), where companies enjoy the benefits (faster genetic gains) and suffer the costs (decreased fertility caused by the increased frequency of deleterious mutations) from changes in the size of their associated network. In this case, the pedigree (relatedness) is the force that binds all the nodes in the network, where two animals are part of the same network (line) if they descend from the same ancestor.

A key element of our analysis is the existence of what we call "supersires", high-prestige bulls who became the founders of lines (families). The more genetics companies breed the same lines, the more newborn bulls will be related to a specific ancestor, and the denser their networks will be. In our analysis, a supersire is a bull whose number of male descendants is on the progeny distribution's 95th quantile.

After identifying the founders, we used propensity score matching to identify 20 other animals per sire with similar trait values but did not have such extensive progeny. These bulls are the founders of the alternative lines that constitute our control group. We then construct the pedigrees by identifying all their progeny from 2005 to 2017 to analyze how the introduction of genomic selection has increased inbreeding rates of supersire lines (treatment group) relative to that of alternative sires (control group).

We use Differences-in-differences of inbreeding rate means to show that superstar lines have persistently higher inbreeding rates from 2012 onwards after controlling for their genetic traits. This results from the breeder's preference for more popular lines independent of their productivity levels to gain a higher market share. According to the Holstein Association of the United States data, each additional percent point of inbreeding has a \$23 revenue loss per cow, implying that increased inbreeding leads to a cost per cow that ranges between \$ 22 and \$ 64 per animal and year. We can then multiply this value by the total number of cows in the US, 89.3 million animals² to get a total cost to the dairy industry that ranges from \$ 1.96 to \$ 5.7 billion for the entire US dairy industry.

²<https://www.nass.usda.gov/Newsroom/2023/01-31-2023.php>

2 Literature Review

~~Selective breeding has been practiced since prehistoric times. However, the discovery of inheritance mechanisms and genetics in the 19th and 20th centuries allowed breeders to identify the link between desired traits in animals and their inheritance mechanisms (Lush, 1937). From an economic standpoint,~~ An animal breeder supplies a product (animal genetics), an inseparable bundle of traits embodied in a single animal. To produce such an animal, the breeder combines a set of inputs that can be split into two groups: a series of inputs exogenously supplied by the producer (labor, feed, capital, veterinary services) and a series of inputs embodied in the animal (traits.) Those inputs are combined on a production function subject to the prices of exogenous inputs (observed) and genetic inputs, whose prices are implicit but can be estimated based on the price paid for a bull's genetics using the hedonic approach (Schroeder et al. 1992 Sy et al. 1997).

Melton et al. (1994, 1979) add an additional element: breeders are paid based on the (positive) deviation of the trait levels of their bulls relative to the population average on a given period. Breeders compete with each other to maximize profits; to do so, they will attempt to release into the market a bull with the highest possible trait values. Dairy farmers demand genetics to improve their productivity in the next period by choosing the bulls with the best characteristics. However, the traits that constitute an animal are marketed as a "bundle" of traits embedded in each animal. In other words, dairy farmers are only interested in the productivity of their cows because their profits depend on it; they are not directly interested in how inbred their cows are; therefore, inbreeding rates do not have a price. *they care because it erodes productivity*

~~We have overlooked one key element of genetic change: its dynamics.~~ Breeders are paid according to their animal's contribution to the genetic improvement of dairy farmers' herds and the expected change in hereditary traits across one generation. This rate of genetic change is affected by several factors, specifically: heritability, selection intensity, and generation interval (Kerr 1984).

$$\text{genetic improvement} = \frac{\text{heritability} \times \text{selection intensity} \times \text{trait variance}}{\text{generation interval}} \quad (1)$$

Heritability is a population parameter specific to each trait (Bourdon, 2000 p. 165) that measures the strength of the relationship between the genetic traits of parents and their offspring, and so is the standard deviation of the observed trait values. **Selection intensity** is the differential

in average trait values between selected animals and the general population at a given time.

Selection intensity measures how "choosy" breeders are in deciding which individuals are selected; more intensity of selection means that only the "best" animals (according to some criterion) are selected to have offspring. If animals were randomly selected, those magnitudes would not differ. The generation interval is the time required to replace one generation with the next; the shorter the interval, the faster the genetic change. Genetic improvement is driven by two forces: selection intensity and generation interval. An increase in selectivity or a decrease in the generation interval increases genetic gains per generation.

A final element in our analysis is the main consequence of increased selection intensity: the mating of close relatives. **Inbreeding** is the mating of individuals more closely related to each other than the average for the population (Bourdon, 2000, p. 337). In the case of dairy cattle, the objective of mating a bull with a cow that is, to a degree, related is to preserve some of the characteristics of the bull or to help create a certain distance to a particular ancestor.

Inbreeding has two consequences: first, it increases the likelihood that an animal's offspring will perform similarly to its ancestors; for example, if a particularly tall bull is mated with a related cow, its offspring will likely be taller than average because the genes that carry such trait are likely to be present in both parents. Second, it favors the expression of deleterious recessive alleles (Charlesworth and Willis, 2009). Since an animal carries two copies of a recessive gene from each parent, it is more likely to express any such traits than an outbred animal.

3 Background

The demand for cattle genetics is made up of all dairy farmers in the US and abroad who periodically buy bull semen to impregnate their cows for two reasons: a cow needs to be pregnant or weaning her calf to produce milk, and also because they are going to need new cows in the upcoming years; a cow's productive life is about four years. Dairy farmers rarely keep bulls within their herds because they do not contribute to their profits, so genetics companies provide the semen to impregnate cows. On the other hand, the supply of dairy genetics is constituted by genetics companies that breed bulls to meet the demand for specific traits. To do so, they choose sires and dams with a particular objective. By doing so, they ensure that the next generation of bulls will have a similar or higher level of traits.

Selective breeding is the only way to carry out directional genetic change (Melton et al., 1979);

by doing so, breeders decide which bulls mate with a specific cow to produce the next generation of animals whose semen will be marketed to dairy farmers, who are interested only in cows, since every all production traits (e.g., milk and protein content) are expressed in females only.

The bovine genome was sequenced in 2008 (Elsik et al., 2009), providing animal breeders with the last missing piece required to make genomic selection commercially viable; the "50K" chip was released that same year, a commercial system that simultaneously detects up to 50,000 SNPs, thus kickstarting the genomic selection era. Genomic testing of young bulls immediately decreased the generation interval (average age of parents when their select offspring is born). It accelerated the genetic gains (variation in breeding values per year) for North America's dairy cattle population (Guinan et al., 2023).

Larger genetics companies adopted genomic selection techniques faster because of their high sunk costs. To correctly identify which gene combinations are correlated with specific traits, a significantly large number of animals had to be sequenced first. According to (VanRaden et al., 2009), a first batch of 3576 Holstein bulls was first sequenced in 2008 and used as the "train set" to identify and predict traits for other 1759 bulls that constituted the "test set" for the statistical methods used to capture the relationship between genes and observed traits.

Increased competition for genetic improvement implies more competition for "superstar" lines, that is, bulls descended from a notorious ancestor (e.g., bulls that fathered many daughters). The more related two animals are, the more likely their offspring is to preserve their genetic traits, but also, they will "inherit" their ancestor's prestige. Higher inbreeding levels lead to a higher likelihood of observing **inbreeding depression**, a decrease in the performance of inbred animals due to lower fertility, higher incidence of rare diseases, and reduced longevity.

The link between genetic improvement and inbreeding rates can be shown by looking at the **Total Performance Index (TPI)**, a selection index used by the Holstein Association of the United States of America. Figure 2 shows a change in the trend starting in 2010 but leveling after 2018.

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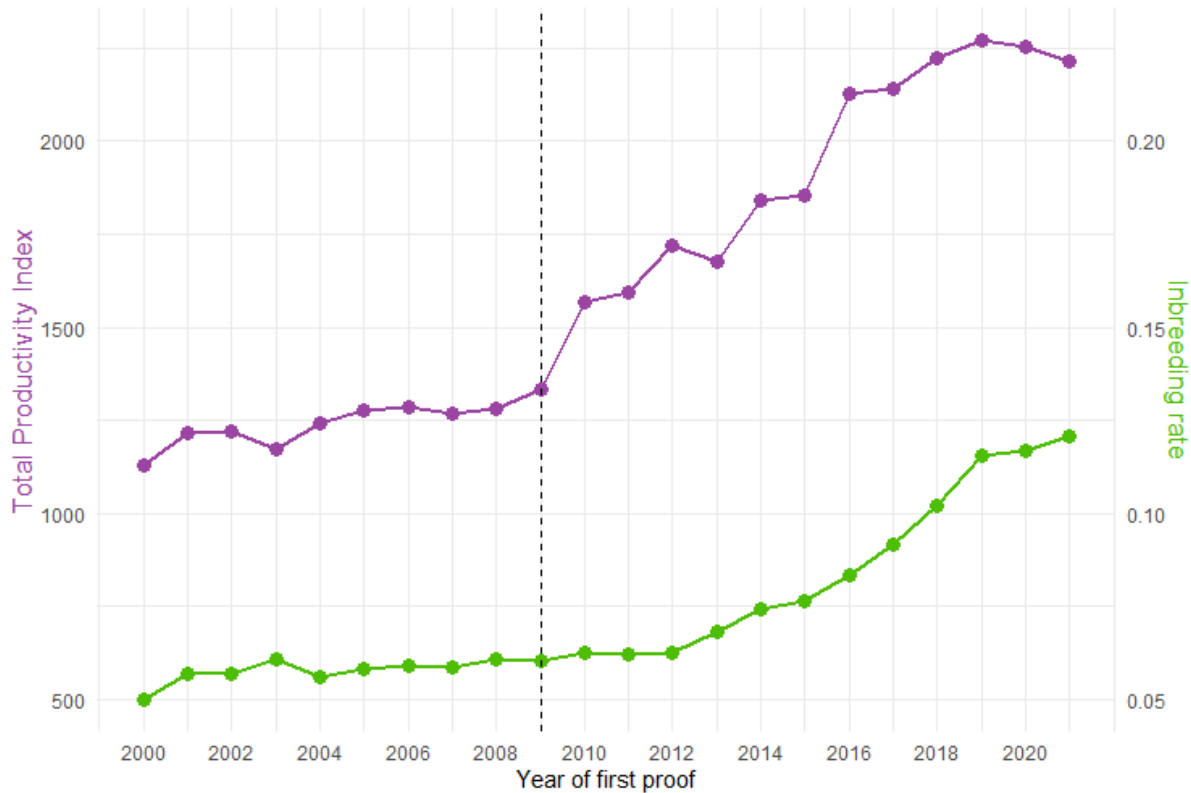


Figure 2: Average Total Performance Index of Holstein bulls
Source: NAAB

With genomic testing, it is no longer necessary to wait for the bull's progeny to increase in size since an animal can be tested as soon as it is born for the presence of specific traits on its genome. Consequently, the generation interval, the average age of parents when offspring are born (Wiggans and Carrillo, 2022), has decreased from around 5.5 years to less than two years, close to the biological minimum.

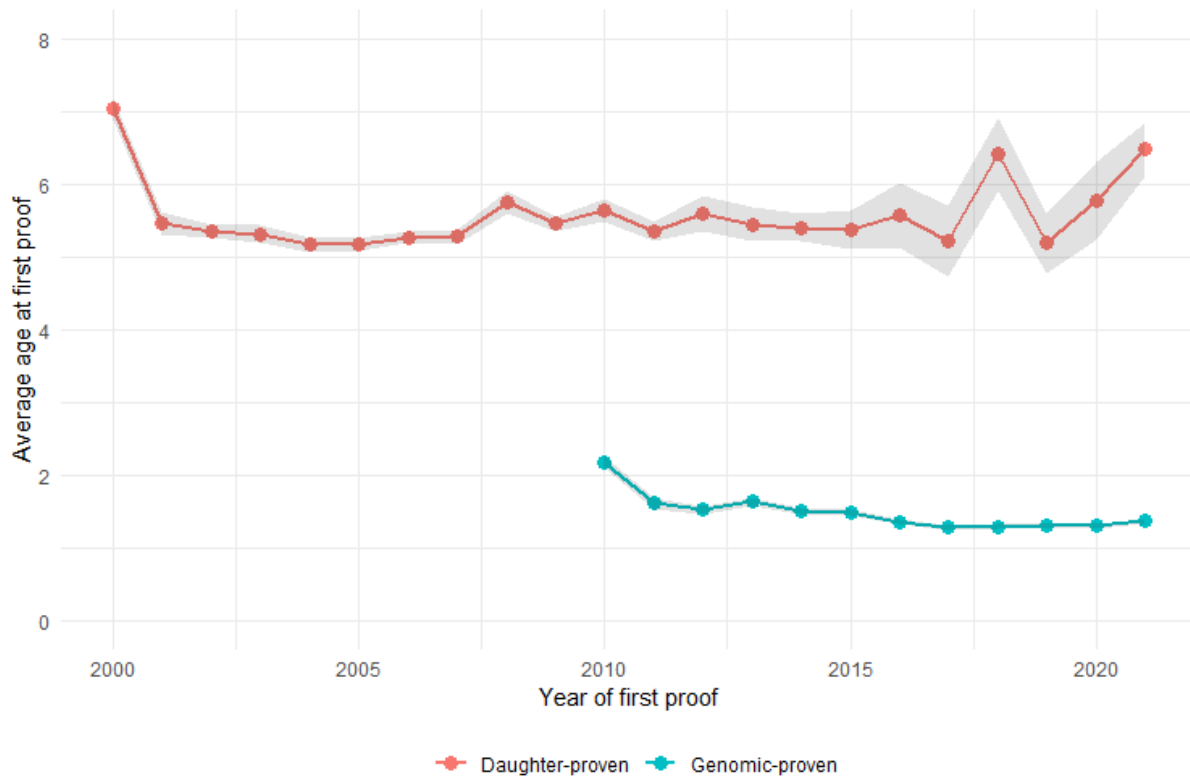


Figure 3: Average age at first proof for Holstein bulls
Source: NAAB

The most important consequence of this shorter generation interval is the increase in genetic progress. According to [García-Ruiz et al. \(2016\)](#), the most significant genetic gains were observed in lowly heritable traits such as DPR (Daughter Pregnancy Rate), PL (Productive Life), and SCS (Somatic Cell Score). Note that Figure 3 also shows that the number of "Active" bulls has steadily decreased, and the wider confidence intervals for the average age reflect that.

4 Data Description

A genetic trait is the portion of an observed value of an animal's characteristics that can be measured directly. In the United States, the National Dairy Herd Improvement Program (NDHIP) collects, manages, and analyzes data from dairy cattle ([Hutchins and Hueth 2023](#)). The information it collects assesses the industry's productivity and estimates the paternal contribution to these traits. The raw data collected from farms is used to estimate a statistical model, called the "animal model" ([Henderson, 1975](#) [Wiggans and VanRaden, 1989](#)), to predict the genetic contribution of an animal as a sire (or dam) after controlling for environmental variables herd, age, pedigree, and season effects. The predicted values for each trait from this model are commonly

called ^{the} predicted transmitting ability (PTA) of a trait.

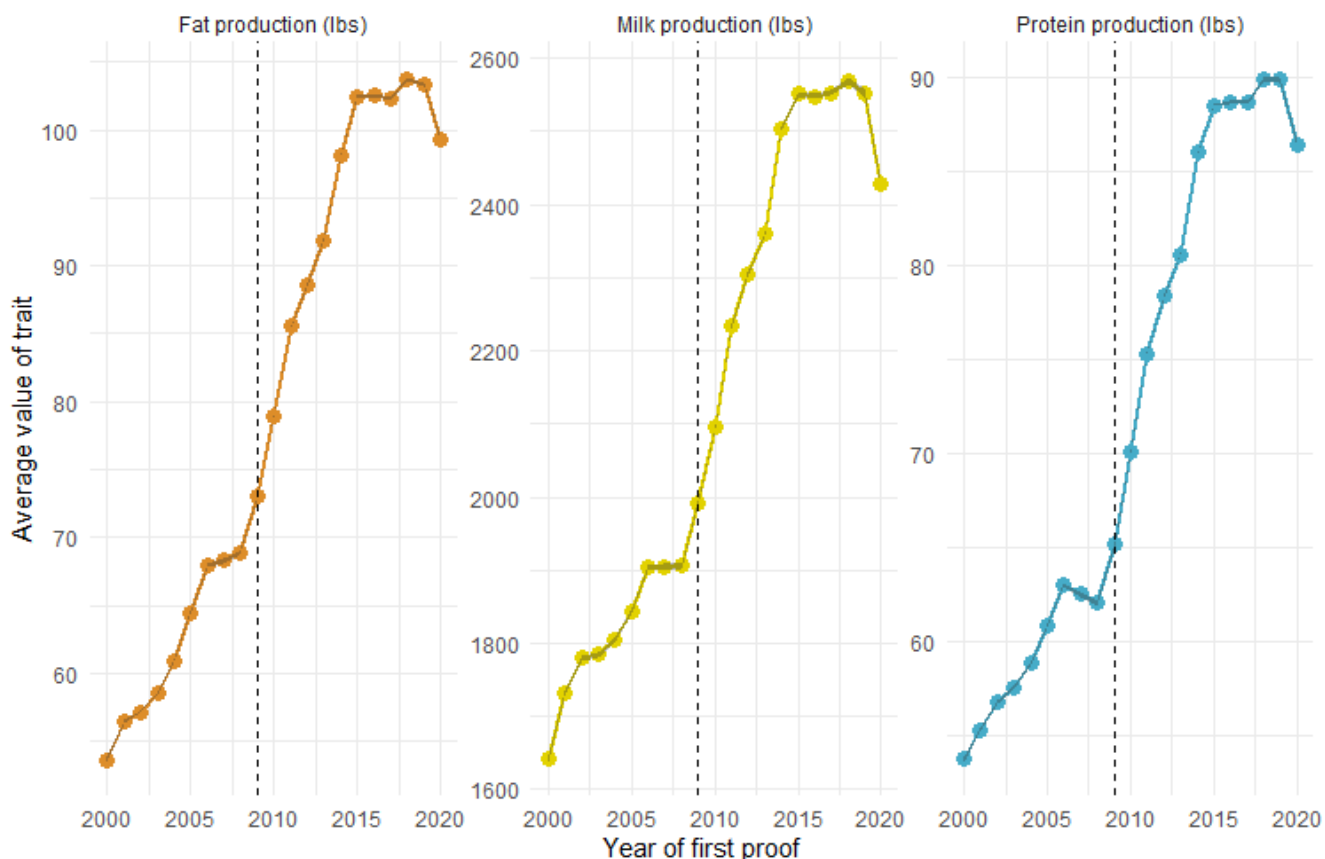


Figure 4: PTAs of production traits
Source: NAAB

The supply of dairy genetics comprises animal breeders and genetics companies; these companies sell semen to dairy farmers, who, in turn, use it to impregnate their cows. Animal breeders and genetics companies compete to offer an animal with the most significant possible improvement over the cohort's average. As a consequence, firms will try to produce bulls with the highest productivity traits; similarly, this will lead to higher rates of inbreeding since the best way to ensure high productivity is through linebreeding, that is, by building a pedigree that has a relatively stable degree of relationship relative to a prestigious ancestor, a "supersire".

However, the more inbred selected animals are, the higher the average inbreeding rate for the entire population in any cohort, thus making inbreeding depression more likely within a herd. The effective population size³ of bulls is at a historic low; according to Mekanjuola et al. (2020), the effective population size of Holstein bulls ranges from 43 to 66 animals. One way to tackle this problem is by importing foreign bulls, which can diversify the gene pool.

³The number of individuals that an idealized population would need to have for inbreeding rates to be the same as in the actual population.

Genomic testing allows genetic companies to detect suitable bulls earlier; thus, they can release those animals into the market as soon as they reach sexual maturity. Figure 5 shows the total number of new bulls released in the market by year and type of firm. The first official genomic results were released in January 2009 for Holsteins and Jerseys (Wiggins and Carrillo 2022). However, not all genetic companies adopted the technology simultaneously.

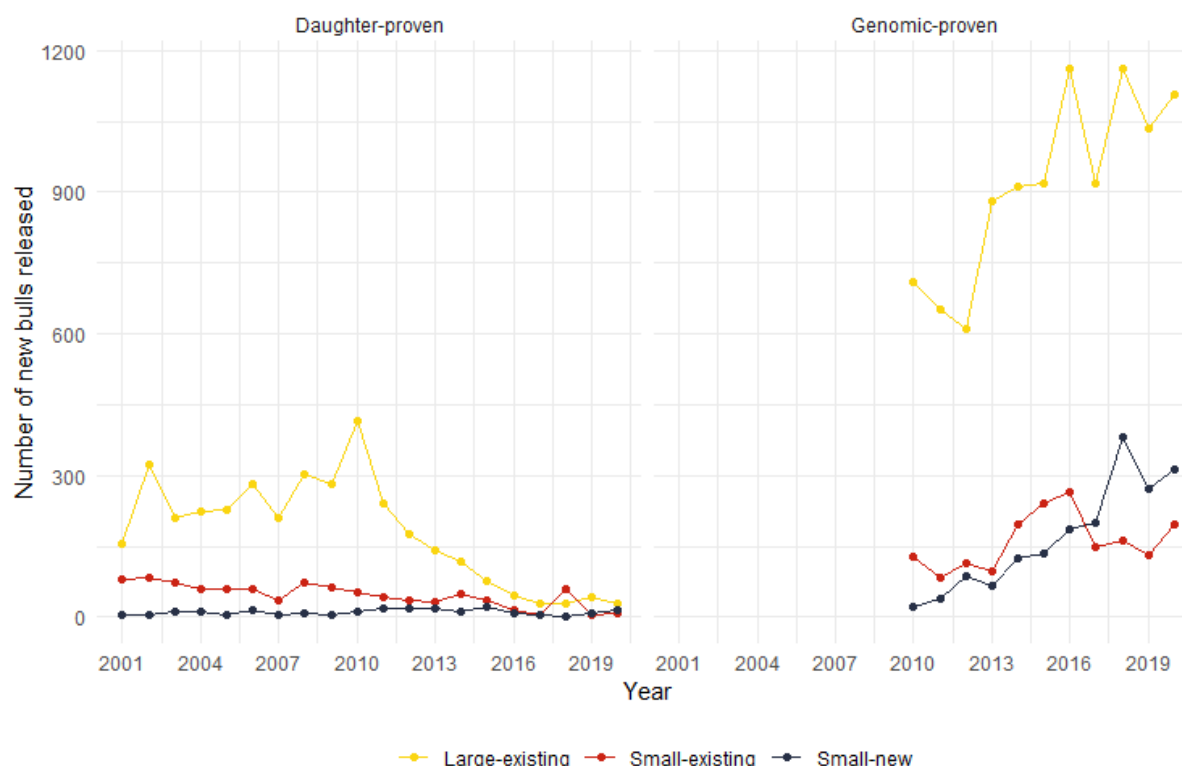


Figure 5: Number of genotyped bulls by firm size
Source: NAAB

We call "large" companies those with the most significant average number of bulls across the entire 2000-2020 period in the North American genetics market. Similarly, a firm is called "preexisting" if it sold bull genetics before 2009; otherwise, they are considered "new."

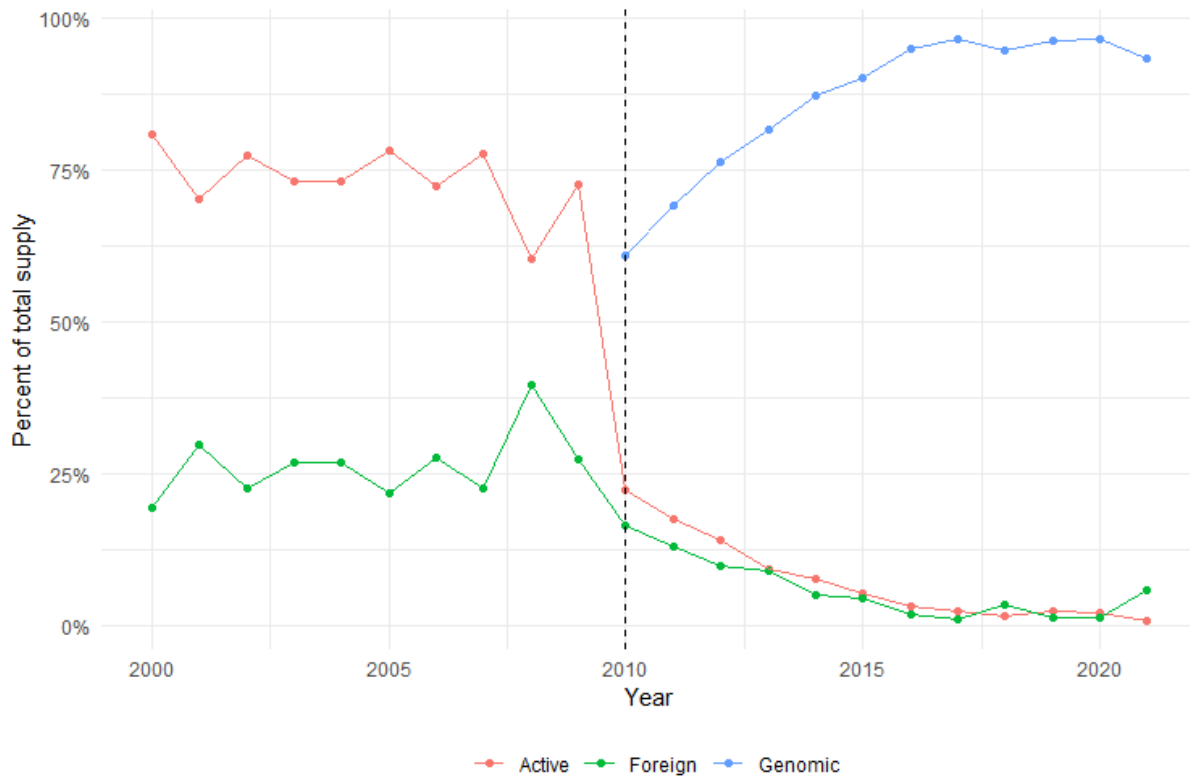


Figure 6: Share of each status of bull by year

In the cattle genetics industry, genomic testing has been adopted remarkably fast. Figure 6 shows how fast genomic testing was adopted in our sample. The high starting costs for genotyping help explain this since large companies were the ones that could initially afford it. Still, more companies have genotyped their calves since genomic selection was released. Figure 7 breaks down the adoption percent by firm size; larger firms genotyped a more significant percent of their newborn calves in 2010, and the share has grown steadily, while smaller firms followed them at a slower rate.

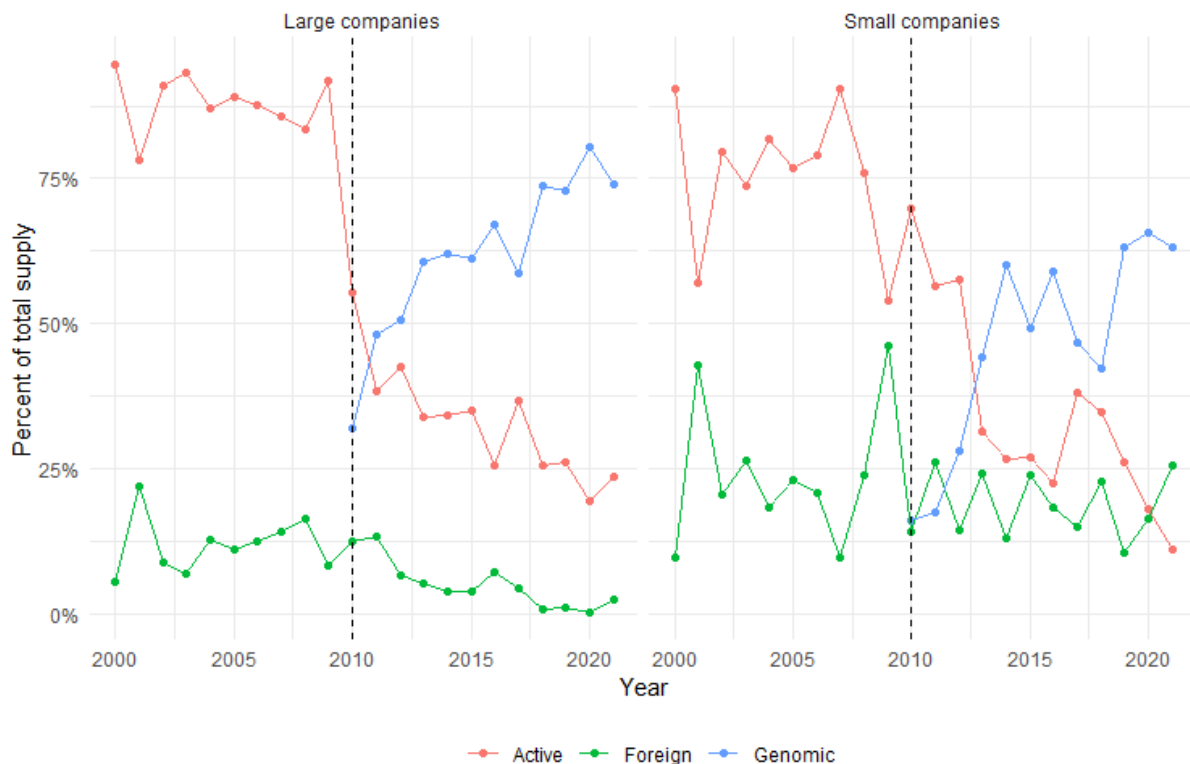


Figure 7: Share of each status of bull by year and company size

Starting in 2010, large firms started genotyping their bulls, and consequently, the number of new bulls increased threefold in that year, when all bulls aged 1 to 4 were released simultaneously. The staggered adoption of genomic selection has further consolidated the market and has also led to more competition for "superstar" lines of bulls. However, the more inbred selected animals are, by definition, also increases the inbreeding of an average animal in any cohort, thus making inbreeding depression more likely within a herd. Figure 5 shows the total number of new bulls released in the market by year and type of firm. We call "large" companies those with the largest share of the North American genetics market. Similarly, a firm is called "preexisting" if it participated in the market of bull genetics before 2009; otherwise, they are considered "new."

5 Empirical Framework

We want to test the hypothesis that the introduction of genomic selection led to higher inbreeding rates because of increased demand for well-known pedigrees fueled by an increment in the productivity of dairy cattle. We want to know the causal effects of introducing this technology; ideally, we need a treatment group that was affected by genomic selection and a treatment group that was not.

What??

We must identify families whose founders had similar trait levels as superstars, but had less "momentum"; that is, less progeny. This is a comparable group because breeders could have invested in them but did not because they were not demanded as highly as superstars. We can track these lines over time. If their inbreeding rates did not differ substantially from those of superstars, then we could argue that genomic selection allowed breeders to identify new less-inbred bulls independent from their ancestry; ~~we will see that this was not the case; genomic selection led to higher inbreeding rates instead.~~

Figure 6 shows that the pace of adoption was very high from the beginning; this makes it unlikely to use non-genomic proven bulls as a counterfactual, mainly because we don't know what breeders choose not to genotype a small number of bulls after 2009. Similarly, we have very few foreign-born bulls to use as a control group; we also don't know the selection criteria to import genetics from those animals. Both alternative control groups suffer from an additional issue: they will have a very small progeny, thus leaving us with little statistical power to detect any relevant treatment effect. To overcome these issues, we use propensity score matching (Smith and Todd, 2005) to define a control group that resembles the treatment in all observable characteristics (except for our outcome: inbreeding rates.)

We define a **line** as the set of direct (male) descendants from a "superstar" sire, which, in turn, can be described in terms of his number of male descendants; a bull is a superstar if his number of sons is in the 95th percentile of the distribution of progeny. Breeders are under high competitive pressure to supply bulls with the highest PTAs. However, the best way to do so is by linebreeding a family of bulls so that the relative distance to an ancestor remains constant across generations. Steyn et al. (2022) and Steyn et al. (2023) show that younger sires capture a more significant share of present-day population genetic variation; hence genotypes of older sires can be discarded from (genomic) evaluations. As generations begin to accumulate, the contribution of a particular sire decreases (on average) exponentially, so we can disregard the impact of distant ancestors.

We include in the sample all animals whose sires were born between 1997 and 2007, and we call those sires the "founders" of each line. Our definition depends critically on the period when the founders of such lines were born; if we choose an earlier date, we cannot find a statistically significant effect due to the high relatedness of all bulls. The reason behind this is that bulls are advertised only with their sire and grandsire, any ancestry beyond that is not taken into account by dairy farmers. If we choose a later date, both treatment and control groups will have little difference in inbreeding rates because there is not enough time for the control group lines to

develop.

After identifying the sires, we can recursively build their pedigrees by matching bulls with their sires until we find the latest animal in the sample. ^{put at beginning} We want to measure the causal effect of introducing genomic testing in 2009 on the average inbreeding in extant lines; the first problem we must tackle is the nature of a valid comparison group. The introduction of genomic testing caused animal breeders to supply more animals from well-known lines, that is, descended from superstar sires; our control group must be the lines descended from other non-superstar sires.

First, we need to match the founders of such lines with other sires that did not father as many sons but had comparable genetic traits. Second, we find all descendants from both groups and compare the evolution of their inbreeding rates through time. The idea behind this procedure is that when those bulls were in the market, any reasonably similar sire could have been chosen but was not, and thus, can be used to construct a counterfactual line.

For every family, we must construct a counterfactual family with new bulls being born yearly, each with an inbreeding rate that we can average across cohorts. Then, the **treatment group** are all lines (families) of animals comprised of direct descendants of "superstar" sires. Similarly, the **control group** comprises descendants from comparable animals matched to superstar bulls by their genetic trait levels.

Our next objective is to identify the counterfactual lines by matching superstar sires with others that were not as successful but had similar genetic traits. To do so, we use a propensity score matching estimator, let $p(X) = \Pr(T_i = 1/X)$ where T is a dummy equal to 1 if bull i 's number of daughters is on the 95th percentile of the progeny distribution and X_i be a vector comprised of the following:

1. Predicted Transmitting Ability for production traits (milk, fat, and protein), health traits (mastitis, metritis, productive life), and reproductive traits (daughter pregnancy rate, effective calving, gestation length).
2. Dummies for the absence of certain recessive genes (Holstein Foundation 2021) such as *brachyspina*, cholesterol deficiency, and mule foot.
3. Dummies for the presence of genes (polled, red coat color, complex vertebral malformation).
4. Dummies for the presence of haplotypes (specific sequences of DNA at different locations in the chromosome) that affect fertility (Holstein Foundation 2018) when present in homozygous form (likely to be expressed when two carrier animals are mated).

We then estimate the propensity score (Cameron and Trivedi, 2005, p. 873), the probability of being a superstar given the level of PTAs and the presence of the entire set of variables listed above.

The validity of our method relies on a series of assumptions (Heckman et al., 1997):

Assumption 1 (Stable Unit Treatment Value Assumption): $(F^{treatment}, F^{control}) \perp\!\!\!\perp T/X$.

The average inbreeding rates of any two lines do not vary with the treatment assigned to other lines. For each unit, there are no different forms of each treatment level (Imbens and Rubin, 2015), conditional on a set of covariates. This assumption implies that conditional on X , the distribution of inbreeding rates for both groups will be the same.

Figure 10 shows the extent to which the inbreeding rates distribution of our matched sample resembles that of the treatment group. Unlike unmatched sires, their inbreeding distribution has a similar mean and variance conditional on the set of genetic traits.

Assumption 2: $0 < \Pr(T_i = 1/X) < 1$ for all X

This assumption implies that we can define a propensity score for every value in the treatment group. Assumptions 1 and 2 together are the “strong ignorability” assumption in Rosenbaum and Rubin (1983), and they are necessary in conjunction to identify causal parameters. The set of matching variables listed above exhausts the entire set of possible selection criteria for breeders; no unobserved variable can influence mating decisions not listed above. Consequently, there is no room for selection of unobservables in this context.

Figure 11 shows the degree of overlapping between superstars and the remaining bulls; since there is a considerable difference in the size of the groups, superstars make up only a tiny fraction of bulls; however, the overlapping is good in the range 0.05–0.50, but there exist a few “superstar” that had little close matches in the sample.

A central part of our argument is that treatment and control groups should have similar inbreeding levels before genomic selection. To test that assumption, in Table 1, we calculate the difference in averages of treatment and control groups for inbreeding rates and a few selected traits. There is no statistically significant difference for most traits between treatment and control groups, so our matching method ensures that the difference between both groups is due to the “prestige” of superstar lines because the bulls in the treatment group had a larger progeny than those in the control group. Each year these bulls will have a new set of descendants being born, each with its inbreeding rate and genetic traits; if our hypothesis holds, then the inbreeding rates of the descendants of “supersires” should be significantly higher than those of non-superstar

descendants.

We then estimate the differential impact of genomic selection (our treatment) on the two types of lines, those from “supersires” and those from “alternative” sires. Equation 3 includes leads and lags of the treatment to check to which degree a pre-treatment balance exists in both groups. This is not a test for parallel trends on our differences-in-differences specification (Cunningham 2021), but it helps us ascertain how comparable both groups are. Table 1 shows that there is no significant pre-treatment difference in both groups and that coefficients from 2010-2011 are not significant either, which coincides with the values from Figure 2 where there is a significant increment in inbreeding rates starting in 2012.

We observe the average inbreeding rates of the treatment and matched control groups before and after genomic selection was introduced. If we divide the 2005-2017 period into two groups, pre-intervention (2005-2009) and post-intervention (2010-2017), denoted as 1 and 2 and let T_i denote whether a bull belongs to treatment ($T_i = 1$) or control ($T_i = 0$), then:

$$F_{it} = T_i F_{it}^{treat} + (1 - T_i) F_{it}^{control}$$

Our causal estimand of interest is the Average Treatment on the Treated (ATT) in the post-intervention period:

$$ATT_{post} = E \left[F_{post}^{treat} - F_{post}^{control} \middle| T_i = 1 \right]$$

This quantity measures the average causal effect on the treated lines in the post-treatment period (Roth et al. 2023). To correctly identify causal effects, a series of assumptions must hold (Sun and Abraham 2021):

Assumption 3 (Parallel trends): $E \left[F_{post}^{control} - F_{pre}^{control} \middle| T_i = 1 \right] = E \left[F_{post}^{control} - F_{pre}^{control} \middle| T_i = 0 \right]$

We assume that the average inbreeding rates for both groups would have evolved in parallel without introducing genomic selection. This assumption is not testable because we cannot observe untreated inbreeding rates after 2009, but it constrains our model specification to a Two-Way Fixed Effects specification we call Model (1):

$$F_{ist} = treat_i + post_t + \alpha_s + (treat_i * Post_t)\beta + \mathbf{X}_{it}\gamma + \varepsilon_{ist} \quad (2)$$

Assumption 4 (No anticipatory effects): $F_{pre}^{control} = F_{pre}^{treatment}$

This assumption ensures that both groups were comparable before the treatment; inbreeding

rates of superstar lines did not increase before the introduction of genomic selection. We use matching methods to ensure that this condition holds. If superstars had superior traits to our matched control group, then this condition would be violated, and we would not be able to parse out the causal effect of inbreeding from the effect of bull's genetic superiority. Assumptions 3 and 4 ensure that the OLS estimates $\hat{\beta}$ from Equation 2 are consistent and asymptotically valid (Roth et al., 2023).

Assumption 5: Let $W_i = (F_{i,pre}, F_{i,post}, T_i)$ be a vector of outcomes and treatment status for unit i . Then, we observe a sample of N independent and identically distributed random draes $W_i \sim F_W(w)$ for a distribution F satisfying parallel trends.

Under Assumptions 3, 4, and 5: $\sqrt{N}(\hat{\beta} - ATT_{post}) \xrightarrow{D} N(0, \sigma^2)$, therefore we can consistently estimate the variance σ^2 clustering at the line level (Bertrand et al., 2004).

We can also decompose the variations in treatment time since every year, a certain number of bulls will be born from lines in both groups, each with its inbreeding rate that can be averaged across treatment and control groups. Equation 3 shows the complete specification with leads and lags of the treated group. This equation decomposes the estimated ATT relative to the treatment average by birth year. (Sun and Abraham (2021) call this specification "Dynamic specification", However, there is no staggered treatment since genomic selection affects all lines in the same year, but inbreeding rates evolve independently after the treatment.

This assumption states that the treatment status of an individual from any given line is not affected by the treatment status of others. In our setting, the treatment (genomic selection) affects all individuals, so there is no selection into treatment. To measure the effects of genomic selection on inbreeding, we estimate the differences-in-differences model:

$$F_{ist} = treat_i * \sum_{\substack{t=2005 \\ t \neq 2009}}^{2017} \beta_t \mathbf{I}(yob = t) + treat_i + \alpha_t + \alpha_s + \mathbf{X}_{it} \gamma + \varepsilon_{ist} \quad (3)$$

Where F_{ist} is the inbreeding rate of bull i born from a line that belongs to either the treatment or control group born in year t , supplied by firm s , $\mathbf{X}_{it}$ is the set of genetic traits, β_t is a year of birth fixed effect, and α_s is a firm fixed effect. Model (2) is a specification that does not contain any covariates (PTAs), and it decomposes the ATT on inbreeding rates by year of birth cohort. Model (3) incorporates PTAs and firm-level fixed effects, and Model (4) includes interactions of PTAs and the post-treatment dummy. We include the last specifications to account for trait-level trends, we also know that the treatment affects both inbreeding rates and PTAs, but we do not have a

structural model for their impact (Lewbel 2019), so we include them in a non-linear fashion as both levels and interactions to reduce the uncertainty about the functional form of our controls X_i .

6 Results

Figure 8 shows the average inbreeding rates and confidence intervals for both treatment and control groups across years. Genomic selection did not increase inbreeding rates immediately; both curves began to diverge substantially after 2011. Inbreeding rates, in turn, started increasing two years later, and both curves diverged substantially from 2012 onwards.

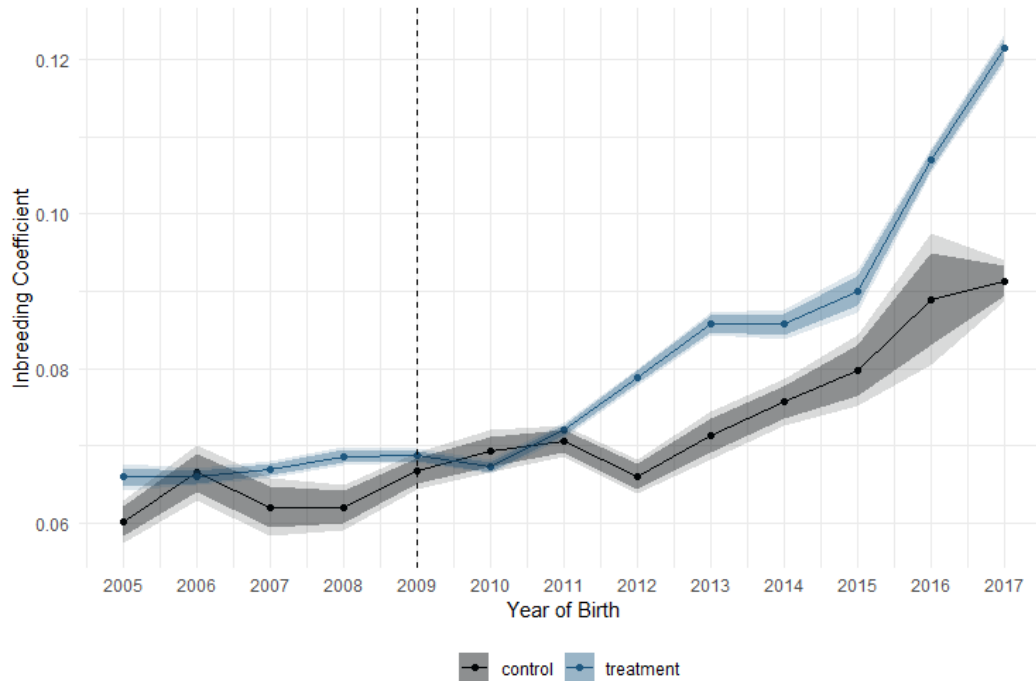


Figure 8: Average inbreeding rates in treatment and control groups

This plot closely matches the evolution of the population average inbreeding rates plotted in Figure 2. This behavior is consistent with our hypothesis that breeders prefer well-known over new lines. Our findings coincide with results reported by Guinan et al. (2023), who also look at the time trends of productivity and reproduction traits in US dairy cattle. The authors also document a decrease in the generation interval and an increase in inbreeding rates of Holstein and Jersey cattle but do not consider the latter to be an issue because the expected inbreeding rates of future progeny adjust genetic trait values.

Table 2 shows the results of three models, model (1) is the analog to Figure 8, including only

the interactions between treatment and year of birth. Model (2) splits the treatment effect by year of birth averages, Model (3) includes the complete set of genetic traits and firm level fixed effects, and Model (4) is the model with traits and a post-treatment dummy. In all specifications, standard errors are clustered at the line level to better deal with the inconsistency generated by the autocorrelation of inbreeding rates between different generations (Bertrand et al. 2004).

Model (1) is the simple differences-in-differences estimation where $post_t$ is a dummy equal to 1 if the bull was born after 2009. The results show that bulls that descend from supersires increased their inbreeding rates 1.3 percent points more than those who descend from less popular bulls, on average. Since the impact of genomic testing on inbreeding rates only became noticeable after 2011, the coefficient underestimates the ATT of genomic selection. Model (2) shows that the treatment-year of birth interactions are positive and significant from 2012 onwards, with the 2017 effect being the highest.

Models (3) and (4) in Table 2 show that there is a positive and significant incremental variation for the treated versus control groups in years 2012-2017; however, the addition of controls in either levels or interactions attenuates the coefficients for 2014 and 2015 to the point of reducing the significance of interactions between treatment and year of birth. Still, in all instances, the coefficients are positive. Regression models show a positive and increasing inbreeding rate in the treatment group comprised of all descendants from prestige sires relative to the control group of descendants of comparable non-superstar sires.

The last period in the sample is 2017 instead of 2020 because we cannot make inferences after 2018 due to a reduction in the number of animals in the control sample, meaning that the number of comparable descendants is too small for us to consider them a valid comparison group. This drastic reduction is a consequence of lower fertility rates in the control group; since those sires fathered fewer sons, their sons will father even fewer grandsons, and after a few years, this group collapses.

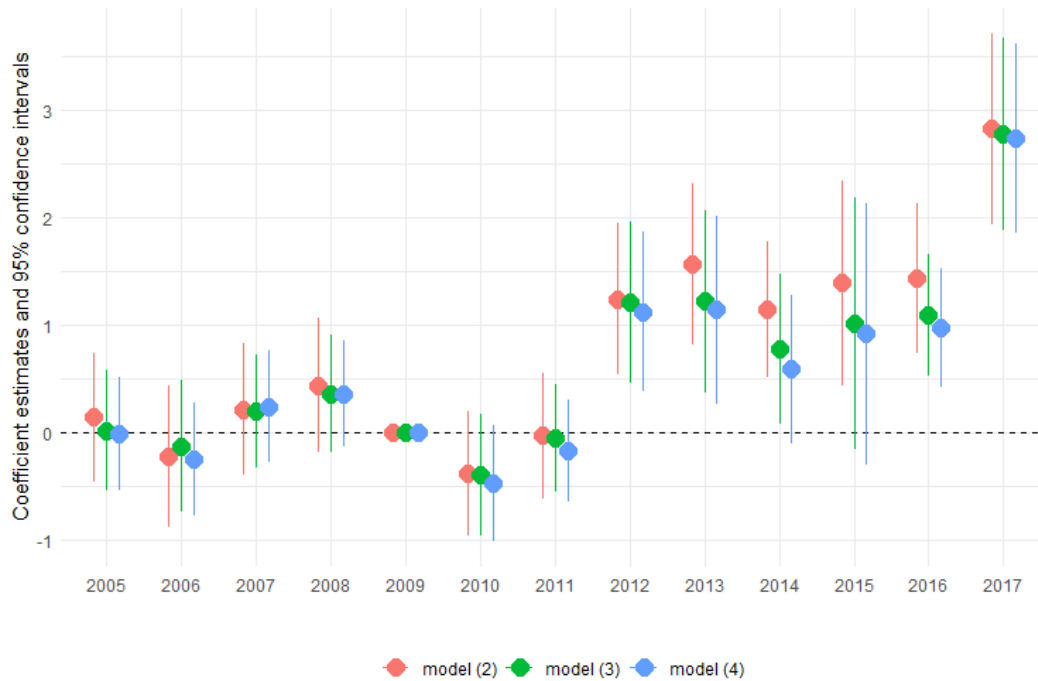


Figure 9: Coefficient plots for different specifications of Equation 3

Figure 9 shows the coefficients from the interaction of year fixed effects and treatment status to assess the impact of inbreeding per year correctly; every year after 2012, the inbreeding rate of the treatment group is between 1 and 2.8 percent points higher than the control group. The coefficient values decrease slightly after controlling for their genetic traits but remain statistically significant. The figure also shows that all pre-treatment coefficients are not significant; we interpret this as evidence in favor of the reasonability of the research design.

7 Discussion

This article investigates the consequences of genomic selection on the American dairy industry. On one side, this new technology significantly improved productivity, but on the other side, it increased the average inbreeding rate of all bulls in the market.

Such increment in productivity was fueled by two effects: a decreased generation interval motivated by a new technology that accurately estimates a bull's genetic traits as soon as it is born; and a more intense competition between animal breeders to produce high-performing bulls. Faster access to information allowed breeders to select better bulls faster, and consequently, genetic gains per generation are now much higher than they used to be in the past.

The most popular method to breed highly productive bulls is by breeding lines that descend

from a famous ancestor; we call these animals “supersires,” and we show that breeding on these well-known lines is the most critical driver of a greater degree of inbreeding among bulls. As the average inbreeding rate increases, every new generation of animals experiences an increased risk of experiencing inbreeding depression, which in turn leads to lower fertility and a higher prevalence of genetic diseases.

Inbreeding depression is a cost that breeders impose on others, a negative externality that is a function of the number of breeders that release closely related bulls into the market. Since pedigrees transmit the externality, it is a network externality that affects all breeders and dairy farmers.

It is impossible to directly test the impact of genomic selection on inbreeding rates because there is no natural control group and no lines that have not been affected by genomic selection. We used a propensity score matching method to find equivalent families based on the founder’s genetic traits but had less progeny, that is, breeders could have invested in them but did not do so. We then track those families across years and observe the average inbreeding rates of their progeny.

We show that bulls from the treatment group (those who descend from popular ancestors) have consistently higher inbreeding rates than those from less-known ancestors from 2012 to 2017. Our analysis shows that the cost of this increased inbreeding ranges between 2 to 5.5 billion dollars.

Inbreeding rates can be reduced significantly by importing bulls from other countries that are not (or less) related to “supersires”. Another possible solution is to adjust the values of genetic traits by inbreeding; the Holstein Association of the United States adjusts the values of PTAs according to inbreeding levels to penalize breeders that select bulls with high inbreeding rates to account for the higher reproduction costs of such animals. The value of this penalty is a crucial element in reducing inbreeding by incentivizing dairy farmers and animal breeders to breed lines that are less related on average by, for example, importing bulls from abroad or by crossbreeding Holsteins with other high-yield milk breeds.

8 Appendix

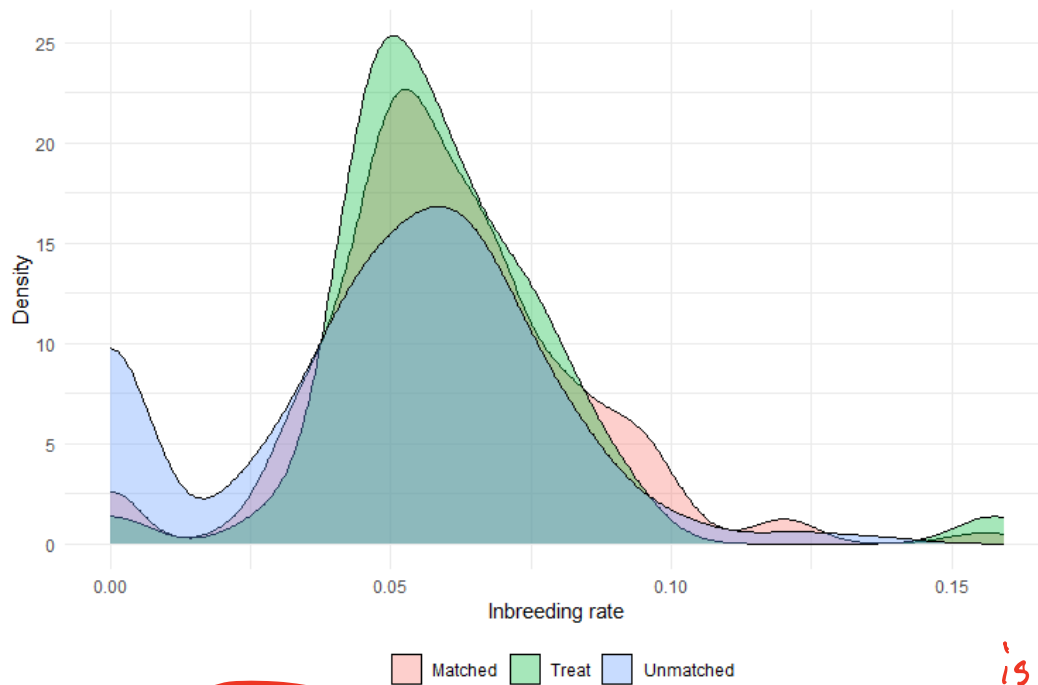


Figure 10: Distribution of Inbreeding rates across groups

*is this
before
treatment?*

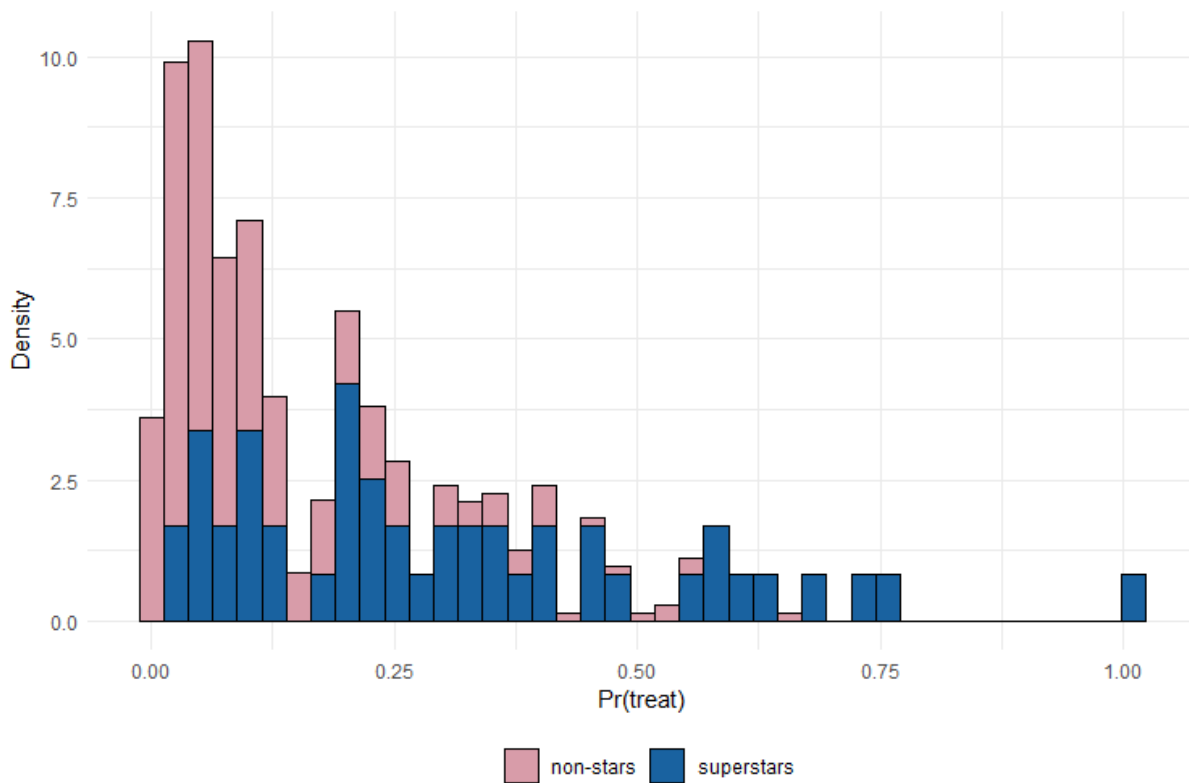


Figure 11: Propensity score by class of bull

yes!
Table 1: Balance table for outcome and matched variables

Variable	Control mean (n = 224)	Treatment mean (n = 43)	t.statistic	p.value	df
Inbreeding rate	2.968	2.904	0.426	0.672	58.147
Milk	0.059	0.059	-0.036	0.971	59.956
Fat	-0.122	-0.170	0.346	0.730	76.548
Protein	-0.851	-1.009	0.416	0.679	61.059
Somatic cell score	-0.151	-0.037	-0.675	0.502	66.764
Productive life	-0.619	-0.677	0.173	0.864	59.829
Daughter preg. rate	-1.224	-0.489	-1.336	0.186	70.101
Heifer conc. rate	-11.953	-5.553	-1.807	0.075	71.841
Cow conc. rate	-0.420	-0.209	-1.278	0.206	65.480
Livability	-0.393	-0.177	-1.816	0.074	70.905
Type	-0.827	-0.551	-1.721	0.090	69.510
Gest. length	0.342	-0.055	1.719	0.090	64.087
Heifer liv.	-0.569	-0.579	0.032	0.975	58.933
Eff. calving	0.101	0.232	-1.549	0.127	58.339
Mastitis	-0.888	-0.764	-0.288	0.774	58.621
Metritis	-0.840	-0.891	0.202	0.840	62.094
Strength	-0.589	-0.581	-0.047	0.963	57.734
Rear legs rear view	-191.265	-24.936	1.581	0.119	67.713
Rear legs side view	-0.027	-0.013	-0.495	0.623	60.964
Foot leg score	-1.348	-1.123	-0.666	0.508	59.099
Teat rear place	-8.284	-3.255	-1.859	0.067	68.421
Rump angle	-0.360	-0.136	-1.447	0.153	64.334
Thurl width	-0.010	0.050	-0.342	0.734	63.537
Foot angle	-0.767	-0.417	-2.044	0.045	68.106
Fore udder	-0.743	-0.432	-1.735	0.087	66.269
Rear udder height	0.142	0.193	-0.254	0.800	60.330
Rear udder width	3.044	2.995	0.690	0.493	52.692
Udder cleft	-0.386	-0.349	-0.235	0.815	77.188
Udder depth	-0.206	-0.267	0.409	0.684	67.108
Teat fron place	-0.343	-0.195	-1.006	0.318	66.497
Teat length	0.197	0.159	0.195	0.846	59.002
Milk fever	-0.312	-0.140	-0.948	0.347	62.811
Stature	-0.255	-0.233	-0.138	0.890	72.664
Dairy form	-0.573	-0.403	-1.442	0.153	73.087
Body depth	-0.401	-0.193	-1.127	0.264	61.777
Sire calv. ease	-0.738	-0.589	-0.989	0.326	73.000
Daughter calv. ease	2.427	2.287	1.188	0.239	63.963

Stars?

Table 2: Difference-in-Differences estimates from Equation 3

	Inbreeding rate (%)			
	No covariates	No covariates	Traits	Traits and interactions
	(1)	(2)	(3)	(4)
<i>treat</i>	0.150 (0.133)	0.074 (0.231)	0.008 (0.218)	0.064 (0.175)
<i>treat</i> × <i>post_2009</i>	1.268*** (0.357)			
<i>treat</i> × <i>I</i> (<i>yob</i> = 2005)		0.134 (0.303)	0.013 (0.287)	−0.022 (0.270)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2006)		−0.231 (0.337)	−0.132 (0.312)	−0.253 (0.269)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2007)		0.212 (0.313)	0.194 (0.270)	0.238 (0.265)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2008)		0.435 (0.318)	0.354 (0.281)	0.355 (0.250)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2010)		−0.390 (0.294)	−0.399 (0.291)	−0.480* (0.275)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2011)		−0.035 (0.297)	−0.063 (0.255)	−0.177 (0.242)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2012)		1.236*** (0.359)	1.205*** (0.385)	1.115*** (0.379)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2013)		1.564*** (0.382)	1.213*** (0.435)	1.137** (0.446)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2014)		1.144*** (0.323)	0.773** (0.354)	0.583* (0.351)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2015)		1.387*** (0.486)	1.012* (0.596)	0.910 (0.620)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2016)		1.429*** (0.359)	1.082*** (0.289)	0.968*** (0.282)
<i>treat</i> × <i>I</i> (<i>yob</i> = 2017)		2.821*** (0.453)	2.770*** (0.455)	2.732*** (0.447)
Observations	15,371	15,371	15,371	15,371
R ²	0.152	0.426	0.453	0.459
Adjusted R ²	0.152	0.425	0.451	0.456
Residual Std. Error	2.262 (df = 15367)	1.864 (df = 15345)	1.821 (df = 15306)	1.813 (df = 15284)

Notes: ***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

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$13/36 \approx 36\% < 90\%$ + have
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